



Syllabus: Connectivity, Matching, Coloring.

Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximum
1 Mark Count	0	0	1	1	0	1	1	0	0	0.5	1
2 Marks Count	0	0	1	2	1	3	1	0	0	1	3
Total Marks	0	0	3	5	2	7	3	0	0	2.5	7

2.1 Counting (3)

2.1.1 Counting: GATE CSE 2001 | Question: 2.15



How many undirected graphs (not necessarily connected) can be constructed out of a given set $V = \{v_1, v_2, \dots, v_n\}$ of n vertices?

- A. $\frac{n(n-1)}{2}$ B. 2^n C. $n!$ D. $2^{\frac{n(n-1)}{2}}$

gatecse-2001 graph-theory normal counting

Answer key

2.1.2 Counting: GATE CSE 2004 | Question: 79



How many graphs on n labeled vertices exist which have at least $\frac{(n^2-3n)}{2}$ edges ?

- A. $\binom{\frac{n^2-n}{2}}{C_{\left(\frac{n^2-3n}{2}\right)}}$ B. $\sum_{k=0}^{\left(\frac{n^2-3n}{2}\right)} \cdot (n^2-n) C_k$
 C. $\binom{\frac{n^2-n}{2}}{C_n}$ D. $\sum_{k=0}^n \cdot \binom{\frac{n^2-n}{2}}{C_k}$

gatecse-2004 graph-theory combinatory normal counting

Answer key

2.1.3 Counting: GATE CSE 2012 | Question: 38



Let G be a complete undirected graph on 6 vertices. If vertices of G are labeled, then the number of distinct cycles of length 4 in G is equal to

- A. 15 B. 30 C. 90 D. 360

gatecse-2012 graph-theory normal marks-to-all counting

Answer key

2.2 Degree of Graph (12)

2.2.1 Degree of Graph: GATE CSE 1987 | Question: 9c



Show that the number of odd-degree vertices in a finite graph is even.

gate1987 graph-theory degree-of-graph descriptive proof

Answer key

2.2.2 Degree of Graph: GATE CSE 1991 | Question: 16-b



Show that all vertices in an undirected finite graph cannot have distinct degrees, if the graph has at least two vertices.

gate1991 graph-theory degree-of-graph descriptive proof

Answer key

2.2.3 Degree of Graph: GATE CSE 1995 | Question: 24



Prove that in finite graph, the number of vertices of odd degree is always even.

Answer key

2.2.4 Degree of Graph: GATE CSE 2003 | Question: 40



A graph $G = (V, E)$ satisfies $|E| \leq 3|V| - 6$. The min-degree of G is defined as $\min_{v \in V} \{\text{degree}(v)\}$. Therefore, min-degree of G cannot be

- A. 3 B. 4 C. 5 D. 6

gatecse-2003 graph-theory normal degree-of-graph

Answer key

2.2.5 Degree of Graph: GATE CSE 2006 | Question: 71



The 2^n vertices of a graph G corresponds to all subsets of a set of size n , for $n \geq 6$. Two vertices of G are adjacent if and only if the corresponding sets intersect in exactly two elements.

The number of vertices of degree zero in G is:

- A. 1 B. n C. $n + 1$ D. 2^n

gatecse-2006 graph-theory normal degree-of-graph

Answer key

2.2.6 Degree of Graph: GATE CSE 2006 | Question: 72



The 2^n vertices of a graph G corresponds to all subsets of a set of size n , for $n \geq 6$. Two vertices of G are adjacent if and only if the corresponding sets intersect in exactly two elements.

The maximum degree of a vertex in G is:

- A. $\left(\frac{n}{2}\right) \cdot 2^{\frac{n}{2}}$ B. 2^{n-2} C. $2^{n-3} \times 3$ D. 2^{n-1}

gatecse-2006 graph-theory normal degree-of-graph

Answer key

2.2.7 Degree of Graph: GATE CSE 2009 | Question: 3



Which one of the following is **TRUE** for any simple connected undirected graph with more than 2 vertices?

- A. No two vertices have the same degree.
B. At least two vertices have the same degree.
C. At least three vertices have the same degree.
D. All vertices have the same degree.

gatecse-2009 graph-theory normal degree-of-graph

Answer key

2.2.8 Degree of Graph: GATE CSE 2010 | Question: 1



Let $G = (V, E)$ be a graph. Define $\xi(G) = \sum_d i_d * d$, where i_d is the number of vertices of degree d in G . If S and T are two different trees with $\xi(S) = \xi(T)$, then

- A. $|S| = 2|T|$ B. $|S| = |T| - 1$
C. $|S| = |T|$ D. $|S| = |T| + 1$

gatecse-2010 graph-theory normal degree-of-graph

Answer key

2.2.9 Degree of Graph: GATE CSE 2010 | Question: 28



The degree sequence of a simple graph is the sequence of the degrees of the nodes in the graph in decreasing order. Which of the following sequences can not be the degree sequence of any graph?

- I. 7, 6, 5, 4, 4, 3, 2, 1
II. 6, 6, 6, 6, 3, 3, 2, 2
III. 7, 6, 6, 4, 4, 3, 2, 2

IV. 8, 7, 7, 6, 4, 2, 1, 1

A. I and II

B. III and IV

C. IV only

D. II and IV

gatecse-2010 graph-theory degree-of-graph

Answer key

2.2.10 Degree of Graph: GATE CSE 2013 | Question: 25



Which of the following statements is/are TRUE for undirected graphs?

P: Number of odd degree vertices is even.

Q: Sum of degrees of all vertices is even.

A. P only

B. Q only

C. Both P and Q

D. Neither P nor Q

gatecse-2013 graph-theory easy degree-of-graph

Answer key

2.2.11 Degree of Graph: GATE CSE 2014 Set 1 | Question: 52



An ordered n -tuple $(d_1, d_2, \dots, d_n)$ with $d_1 \geq d_2 \geq \dots \geq d_n$ is called *graphic* if there exists a simple undirected graph with n vertices having degrees $d_1, d_2, \dots, d_n$ respectively. Which one of the following 6-tuples is NOT graphic?

A. (1, 1, 1, 1, 1, 1)

B. (2, 2, 2, 2, 2, 2)

C. (3, 3, 3, 1, 0, 0)

D. (3, 2, 1, 1, 1, 0)

gatecse-2014-set1 graph-theory normal degree-of-graph

Answer key

2.2.12 Degree of Graph: GATE CSE 2017 Set 2 | Question: 23



G is an undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is _____.

gatecse-2017-set2 graph-theory numerical-answers degree-of-graph

Answer key

2.3

Graph Coloring (11)

2.3.1 Graph Coloring: GATE CSE 2002 | Question: 1.4



The minimum number of colours required to colour the vertices of a cycle with n nodes in such a way that no two adjacent nodes have the same colour is

A. 2

B. 3

C. 4

D. $n - 2 \lfloor \frac{n}{2} \rfloor + 2$

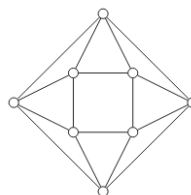
gatecse-2002 graph-theory graph-coloring normal

Answer key

2.3.2 Graph Coloring: GATE CSE 2004 | Question: 77



The minimum number of colours required to colour the following graph, such that no two adjacent vertices are assigned the same color, is



A. 2

B. 3

C. 4

D. 5

gatecse-2004 graph-theory graph-coloring easy

Answer key

2.3.3 Graph Coloring: GATE CSE 2009 | Question: 2



What is the chromatic number of an n vertex simple connected graph which does not contain any odd length cycle? Assume $n > 2$.

- A. 2 B. 3 C. $n - 1$ D. n

gatecse-2009 graph-theory graph-coloring normal

Answer key

2.3.4 Graph Coloring: GATE CSE 2016 Set 2 | Question: 03



The minimum number of colours that is sufficient to vertex-colour any planar graph is _____.

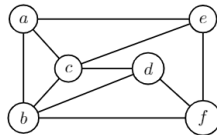
gatecse-2016-set2 graph-theory graph-coloring normal numerical-answers

Answer key

2.3.5 Graph Coloring: GATE CSE 2018 | Question: 18



The chromatic number of the following graph is _____



graph-theory graph-coloring numerical-answers gatecse-2018 one-mark

Answer key

2.3.6 Graph Coloring: GATE CSE 2020 | Question: 52



Graph G is obtained by adding vertex s to $K_{3,4}$ and making s adjacent to every vertex of $K_{3,4}$. The minimum number of colours required to edge-colour G is _____

gatecse-2020 numerical-answers graph-theory graph-coloring two-marks

Answer key

2.3.7 Graph Coloring: GATE CSE 2023 | Question: 45



Let G be a simple, finite, undirected graph with vertex set $\{v_1, \dots, v_n\}$. Let $\Delta(G)$ denote the maximum degree of G and let $\mathbb{N} = \{1, 2, \dots\}$ denote the set of all possible colors. Color the vertices of G using the following greedy strategy: for $i = 1, \dots, n$

$\text{color}(v_i) \leftarrow \min \{j \in \mathbb{N} : \text{no neighbour of } v_i \text{ is colored } j\}$

Which of the following statements is/are TRUE?

- A. This procedure results in a proper vertex coloring of G .
B. The number of colors used is at most $\Delta(G) + 1$.
C. The number of colors used is at most $\Delta(G)$.
D. The number of colors used is equal to the chromatic number of G .

gatecse-2023 graph-theory graph-coloring multiple-selects two-marks

Answer key

2.3.8 Graph Coloring: GATE CSE 2024 | Set 1 | Question: 41



The chromatic number of a graph is the minimum number of colours used in a *proper* colouring of the graph. Let G be any graph with n vertices and chromatic number k . Which of the following statements is/are always TRUE?

- A. G contains a complete subgraph with k vertices
B. G contains an independent set of size at least n/k
C. G contains at least $k(k-1)/2$ edges

D. G contains a vertex of degree at least k

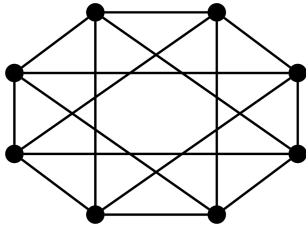
gatecse2024-set1 multiple-selects graph-theory graph-coloring two-marks

Answer key

2.3.9 Graph Coloring: GATE CSE 2024 | Set 2 | Question: 50



The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. The chromatic number of the following graph is _____.



gatecse2024-set2 graph-theory numerical-answers graph-coloring two-marks

Answer key

2.3.10 Graph Coloring: GATE IT 2006 | Question: 25



Consider the undirected graph G defined as follows. The vertices of G are bit strings of length n . We have an edge between vertex u and vertex v if and only if u and v differ in exactly one bit position (in other words, v can be obtained from u by flipping a single bit). The ratio of the chromatic number of G to the diameter of G is,

- A. $\frac{1}{(2^{n-1})}$ B. $(\frac{1}{n})$ C. $(\frac{2}{n})$ D. $(\frac{3}{n})$

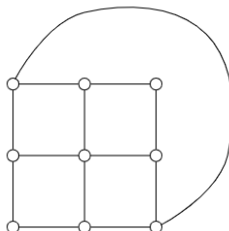
gateit-2006 graph-theory graph-coloring normal

Answer key

2.3.11 Graph Coloring: GATE IT 2008 | Question: 3



What is the chromatic number of the following graph?



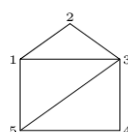
- A. 2 B. 3 C. 4 D. 5

gateit-2008 graph-theory graph-coloring normal

Answer key

2.4 Graph Connectivity (40)

2.4.1 Graph Connectivity: GATE CSE 1987 | Question: 9d



Specify an adjacency-lists representation of the undirected graph given above.

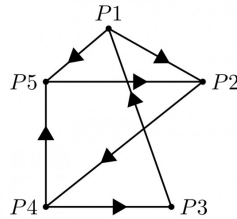
gate1987 graph-theory easy graph-connectivity descriptive

Answer key

2.4.2 Graph Connectivity: GATE CSE 1988 | Question: 2xvi



Write the adjacency matrix representation of the graph given in below figure.



gate1988 descriptive graph-theory graph-connectivity

Answer key

2.4.3 Graph Connectivity: GATE CSE 1990 | Question: 1-viii



A graph which has the same number of edges as its complement must have number of vertices congruent to _____ or _____ modulo 4.

gate1990 graph-theory graph-connectivity fill-in-the-blanks

Answer key

2.4.4 Graph Connectivity: GATE CSE 1991 | Question: 01,xv



The maximum number of possible edges in an undirected graph with n vertices and k components is _____.

gate1991 graph-theory graph-connectivity normal fill-in-the-blanks

Answer key

2.4.5 Graph Connectivity: GATE CSE 1992 | Question: 03,iii



How many edges can there be in a forest with p components having n vertices in all?

gate1992 graph-theory graph-connectivity descriptive

Answer key

2.4.6 Graph Connectivity: GATE CSE 1993 | Question: 8.1



Consider a simple connected graph G with n vertices and n edges ($n > 2$). Then, which of the following statements are true?

- A. G has no cycles
- B. The graph obtained by removing any edge from G is not connected
- C. G has at least one cycle
- D. The graph obtained by removing any two edges from G is not connected
- E. None of the above

gate1993 graph-theory graph-connectivity easy multiple-selects

Answer key

2.4.7 Graph Connectivity: GATE CSE 1994 | Question: 1.6, ISRO2008-29



The number of distinct simple graphs with up to three nodes is

- A. 15
- B. 10
- C. 7
- D. 9

gate1994 graph-theory graph-connectivity combinatory normal isro2008 counting

Answer key

2.4.8 Graph Connectivity: GATE CSE 1994 | Question: 2.5



The number of edges in a regular graph of degree d and n vertices is _____

Answer key

2.4.9 Graph Connectivity: GATE CSE 1994 | Question: 24



An independent set in a graph is a subset of vertices such that no two vertices in the subset are connected by an edge. An incomplete scheme for a greedy algorithm to find a maximum independent set in a tree is given below:

```
V: Set of all vertices in the tree;
I := ∅
while V ≠ ∅ do
begin
  select a vertex u ∈ V such that
  _____;
  V := V - {u};
  if u is such that
  _____ then I := I ∪ {u}
end;
Output(I);
```

- Complete the algorithm by specifying the property of vertex u in each case.
- What is the time complexity of the algorithm?

Answer key

2.4.10 Graph Connectivity: GATE CSE 1995 | Question: 1.25



The minimum number of edges in a connected cyclic graph on n vertices is:

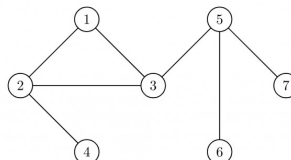
- $n - 1$
- n
- $n + 1$
- None of the above

Answer key

2.4.11 Graph Connectivity: GATE CSE 1999 | Question: 1.15



The number of articulation points of the following graph is



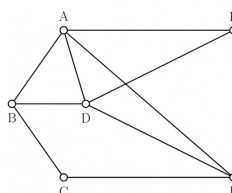
- 0
- 1
- 2
- 3

Answer key

2.4.12 Graph Connectivity: GATE CSE 1999 | Question: 5



Let G be a connected, undirected graph. A cut in G is a set of edges whose removal results in G being broken into two or more components, which are not connected with each other. The size of a cut is called its cardinality. A min-cut of G is a cut in G of minimum cardinality. Consider the following graph:



- Which of the following sets of edges is a cut?
 - $\{(A, B), (E, F), (B, D), (A, E), (A, D)\}$
 - $\{(B, D), (C, F), (A, B)\}$

- b. What is cardinality of min-cut in this graph?
- c. Prove that if a connected undirected graph G with n vertices has a min-cut of cardinality k , then G has at least $\left(\frac{n \times k}{2}\right)$ edges.

gate1999 graph-theory graph-connectivity normal descriptive proof

Answer key

2.4.13 Graph Connectivity: GATE CSE 2002 | Question: 1.25, ISRO2008-30, ISRO2016-6



The maximum number of edges in a n -node undirected graph without self loops is

- A. n^2 B. $\frac{n(n-1)}{2}$ C. $n - 1$ D. $\frac{(n+1)(n)}{2}$

gatecse-2002 graph-theory easy isro2008 isro2016 graph-connectivity

Answer key

2.4.14 Graph Connectivity: GATE CSE 2003 | Question: 8, ISRO2009-53



Let G be an arbitrary graph with n nodes and k components. If a vertex is removed from G , the number of components in the resultant graph must necessarily lie down between

- A. k and n B. $k - 1$ and $k + 1$
C. $k - 1$ and $n - 1$ D. $k + 1$ and $n - k$

gatecse-2003 graph-theory graph-connectivity normal isro2009

Answer key

2.4.15 Graph Connectivity: GATE CSE 2004 | Question: 81



Let $G_1 = (V, E_1)$ and $G_2 = (V, E_2)$ be connected graphs on the same vertex set V with more than two vertices. If $G_1 \cap G_2 = (V, E_1 \cap E_2)$ is not a connected graph, then the graph $G_1 \cup G_2 = (V, E_1 \cup E_2)$

- A. cannot have a cut vertex B. must have a cycle
C. must have a cut-edge (bridge) D. has chromatic number strictly greater than those of G_1 and G_2

gatecse-2004 graph-theory normal graph-connectivity

Answer key

2.4.16 Graph Connectivity: GATE CSE 2005 | Question: 11



Let G be a simple graph with 20 vertices and 100 edges. The size of the minimum vertex cover of G is 8. Then, the size of the maximum independent set of G is:

- A. 12 B. 8 C. less than 8 D. more than 12

gatecse-2005 graph-theory normal graph-connectivity

Answer key

2.4.17 Graph Connectivity: GATE CSE 2006 | Question: 73



The 2^n vertices of a graph G corresponds to all subsets of a set of size n , for $n \geq 6$. Two vertices of G are adjacent if and only if the corresponding sets intersect in exactly two elements.

The number of connected components in G is:

- A. n B. $n + 2$ C. $2^{\frac{n}{2}}$ D. $\frac{2^n}{n}$

gatecse-2006 graph-theory normal graph-connectivity

Answer key

2.4.18 Graph Connectivity: GATE CSE 2007 | Question: 23



Which of the following graphs has an Eulerian circuit?

- A. Any k -regular graph where k is an even number. B. A complete graph on 90 vertices.
C. The complement of a cycle on 25 vertices. D. None of the above

gatecse-2007 graph-theory normal graph-connectivity

Answer key

2.4.19 Graph Connectivity: GATE CSE 2008 | Question: 42



G is a graph on n vertices and $2n - 2$ edges. The edges of G can be partitioned into two edge-disjoint spanning trees. Which of the following is NOT true for G ?

- A. For every subset of k vertices, the induced subgraph has at most $2k - 2$ edges.
- B. The minimum cut in G has at least 2 edges.
- C. There are at least 2 edge-disjoint paths between every pair of vertices.
- D. There are at least 2 vertex-disjoint paths between every pair of vertices.

gatecse-2008 graph-connectivity normal

Answer key

2.4.20 Graph Connectivity: GATE CSE 2013 | Question: 26



The line graph $L(G)$ of a simple graph G is defined as follows:

- There is exactly one vertex $v(e)$ in $L(G)$ for each edge e in G .
- For any two edges e and e' in G , $L(G)$ has an edge between $v(e)$ and $v(e')$, if and only if e and e' are incident with the same vertex in G .

Which of the following statements is/are TRUE?

- (P) The line graph of a cycle is a cycle.
- (Q) The line graph of a clique is a clique.
- (R) The line graph of a planar graph is planar.
- (S) The line graph of a tree is a tree.

A. P only

B. P and R only

C. R only

D. P, Q and S only

gatecse-2013 graph-theory normal graph-connectivity

Answer key

2.4.21 Graph Connectivity: GATE CSE 2014 Set 1 | Question: 51



Consider an undirected graph G where self-loops are not allowed. The vertex set of G is $\{(i, j) \mid 1 \leq i \leq 12, 1 \leq j \leq 12\}$. There is an edge between (a, b) and (c, d) if $|a - c| \leq 1$ and $|b - d| \leq 1$. The number of edges in this graph is _____.

gatecse-2014-set1 graph-theory numerical-answers normal graph-connectivity

Answer key

2.4.22 Graph Connectivity: GATE CSE 2014 Set 2 | Question: 3



The maximum number of edges in a bipartite graph on 12 vertices is _____.

gatecse-2014-set2 graph-theory graph-connectivity numerical-answers normal

Answer key

2.4.23 Graph Connectivity: GATE CSE 2014 Set 3 | Question: 51



If G is the forest with n vertices and k connected components, how many edges does G have?

A. $\lfloor \frac{n}{k} \rfloor$

B. $\lceil \frac{n}{k} \rceil$

C. $n - k$

D. $n - k + 1$

gatecse-2014-set3 graph-theory graph-connectivity normal

Answer key

2.4.24 Graph Connectivity: GATE CSE 2015 Set 2 | Question: 50



In a connected graph, a bridge is an edge whose removal disconnects the graph. Which one of the following statements is true?

- A. A tree has no bridges
- B. A bridge cannot be part of a simple cycle
- C. Every edge of a clique with size ≥ 3 is a bridge (A clique is any complete subgraph of a graph)
- D. A graph with bridges cannot have cycle

gatecse-2015-set2 graph-theory graph-connectivity easy

Answer key

2.4.25 Graph Connectivity: GATE CSE 2019 | Question: 12



Let G be an undirected complete graph on n vertices, where $n > 2$. Then, the number of different Hamiltonian cycles in G is equal to

- A. $n!$
- B. $(n-1)!$
- C. 1
- D. $\frac{(n-1)!}{2}$

gatecse-2019 engineering-mathematics discrete-mathematics graph-theory graph-connectivity one-mark

Answer key

2.4.26 Graph Connectivity: GATE CSE 2019 | Question: 38



Let G be any connected, weighted, undirected graph.

- I. G has a unique minimum spanning tree, if no two edges of G have the same weight.
- II. G has a unique minimum spanning tree, if, for every cut of G , there is a unique minimum-weight edge crossing the cut.

Which of the following statements is/are TRUE?

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

gatecse-2019 engineering-mathematics discrete-mathematics graph-theory graph-connectivity two-marks minimum-spanning-tree algorithms graph-algorithms

Answer key

2.4.27 Graph Connectivity: GATE CSE 2021 Set 1 | Question: 36



Let $G = (V, E)$ be an undirected unweighted connected graph. The *diameter* of G is defined as:

$$\text{diam}(G) = \max_{u, v \in V} \{\text{the length of shortest path between } u \text{ and } v\}$$

Let M be the adjacency matrix of G .

Define graph G_2 on the same set of vertices with adjacency matrix N , where

$$N_{ij} = \begin{cases} 1 & \text{if } M_{ij} > 0 \text{ or } P_{ij} > 0, \text{ where } P = M^2 \\ 0 & \text{otherwise} \end{cases}$$

Which one of the following statements is true?

- A. $\text{diam}(G_2) \leq \lceil \text{diam}(G)/2 \rceil$
- B. $\lceil \text{diam}(G)/2 \rceil < \text{diam}(G_2) < \text{diam}(G)$
- C. $\text{diam}(G_2) = \text{diam}(G)$
- D. $\text{diam}(G) < \text{diam}(G_2) \leq 2 \text{diam}(G)$

gatecse-2021-set1 graph-theory graph-connectivity two-marks

Answer key

2.4.28 Graph Connectivity: GATE CSE 2022 | Question: 20



Consider a simple undirected graph of 10 vertices. If the graph is disconnected, then the maximum number of edges it can have is _____.

gatecse-2022 numerical-answers graph-theory graph-connectivity one-mark

Answer key

2.4.29 Graph Connectivity: GATE CSE 2022 | Question: 27



Consider a simple undirected unweighted graph with at least three vertices. If A is the adjacency matrix of the graph, then the number of 3-cycles in the graph is given by the trace of

- A. A^3
- B. A^3 divided by 2
- C. A^3 divided by 3
- D. A^3 divided by 6

gatecse-2022 graph-theory graph-connectivity two-marks

Answer key

2.4.30 Graph Connectivity: GATE CSE 2022 | Question: 42



Which of the properties hold for the adjacency matrix A of a simple undirected unweighted graph having n vertices?

- A. The diagonal entries of A^2 are the degrees of the vertices of the graph.
- B. If the graph is connected, then none of the entries of $A^{n-1} + I_n$ can be zero.
- C. If the sum of all the elements of A is at most $2(n-1)$, then the graph must be acyclic.
- D. If there is at least a 1 in each of A 's rows and columns, then the graph must be connected.

gatecse-2022 graph-theory graph-connectivity multiple-selects two-marks

Answer key

2.4.31 Graph Connectivity: GATE CSE 2024 | Set 1 | Question: 24



The number of spanning trees in a *complete* graph of 4 vertices labelled A, B, C, and D is _____.

gatecse2024-set1 numerical-answers graph-theory graph-connectivity one-mark

Answer key

2.4.32 Graph Connectivity: GATE CSE 2024 | Set 2 | Question: 41



Let G be an undirected connected graph in which every edge has a positive integer weight. Suppose that every spanning tree in G has *even* weight. Which of the following statements is/are TRUE for *every* such graph G ?

- A. All edges in G have even weight
- B. All edges in G have even weight **OR** all edges in G have odd weight
- C. In each cycle C in G , all edges in C have even weight
- D. In each cycle C in G , either all edges in C have even weight **OR** all edges in C have odd weight

gatecse2024-set2 graph-theory multiple-selects graph-connectivity two-marks

Answer key

2.4.33 Graph Connectivity: GATE CSE 2024 | Set 2 | Question: 7



Let A be the adjacency matrix of a simple undirected graph G . Suppose A is its own inverse. Which one of the following statements is *always* TRUE?

- A. G is a cycle
- B. G is a perfect matching
- C. G is a complete graph
- D. There is no such graph G

gatecse2024-set2 graph-theory graph-connectivity one-mark

Answer key

2.4.34 Graph Connectivity: GATE IT 2004 | Question: 37



What is the number of vertices in an undirected connected graph with 27 edges, 6 vertices of degree 2, 3 vertices of degree 4 and remaining of degree 3?

- A. 10
- B. 11
- C. 18
- D. 19

gateit-2004 graph-theory graph-connectivity normal

Answer key

2.4.35 Graph Connectivity: GATE IT 2004 | Question: 5



What is the maximum number of edges in an acyclic undirected graph with n vertices?

- A. $n - 1$ B. n C. $n + 1$ D. $2n - 1$

gateit-2004 graph-theory graph-connectivity normal

Answer key

2.4.36 Graph Connectivity: GATE IT 2005 | Question: 56



Let G be a directed graph whose vertex set is the set of numbers from 1 to 100. There is an edge from a vertex i to a vertex j iff either $j = i + 1$ or $j = 3i$. The minimum number of edges in a path in G from vertex 1 to vertex 100 is

- A. 4 B. 7 C. 23 D. 99

gateit-2005 graph-theory graph-connectivity normal

Answer key

2.4.37 Graph Connectivity: GATE IT 2006 | Question: 11



If all the edge weights of an undirected graph are positive, then any subset of edges that connects all the vertices and has minimum total weight is a

- A. Hamiltonian cycle B. grid C. hypercube D. tree

gateit-2006 graph-theory graph-connectivity normal

Answer key

2.4.38 Graph Connectivity: GATE IT 2007 | Question: 25



What is the largest integer m such that every simple connected graph with n vertices and n edges contains at least m different spanning trees ?

- A. 1 B. 2 C. 3 D. n

gateit-2007 graph-theory graph-connectivity normal

Answer key

2.4.39 Graph Connectivity: GATE IT 2008 | Question: 27



G is a simple undirected graph. Some vertices of G are of odd degree. Add a node v to G and make it adjacent to each odd degree vertex of G . The resultant graph is sure to be

- A. regular B. complete C. Hamiltonian D. Euler

gateit-2008 graph-theory graph-connectivity normal

Answer key

2.4.40 Graph Connectivity: GATE IT 2008 | Question: 4



What is the size of the smallest MIS (Maximal Independent Set) of a chain of nine nodes?

- A. 5 B. 4 C. 3 D. 2

gateit-2008 normal graph-connectivity

Answer key

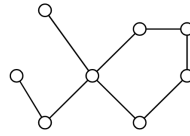
2.5

Graph Isomorphism (3)

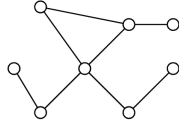
2.5.1 Graph Isomorphism: GATE CSE 2012 | Question: 26



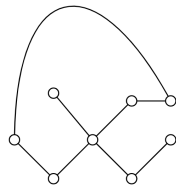
Which of the following graphs is isomorphic to



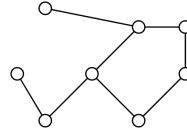
A.



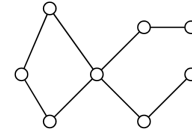
B.



C.



D.



gatecse-2012 graph-theory graph-isomorphism normal

Answer key

2.5.2 Graph Isomorphism: GATE CSE 2014 Set 2 | Question: 51



A cycle on n vertices is isomorphic to its complement. The value of n is _____.

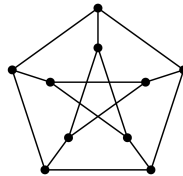
gatecse-2014-set2 graph-theory numerical-answers normal graph-isomorphism

Answer key

2.5.3 Graph Isomorphism: GATE CSE 2022 | Question: 40

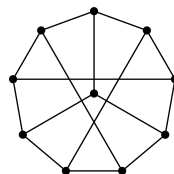


The following simple undirected graph is referred to as the Peterson graph.



Which of the following statements is/are TRUE?

- A. The chromatic number of the graph is 3.
- B. The graph has a Hamiltonian path.
- C. The following graph is isomorphic to the Peterson graph.



- D. The size of the largest independent set of the given graph is 3. (A subset of vertices of a graph form an independent set if no two vertices of the subset are adjacent.)

gatecse-2022 graph-theory graph-isomorphism multiple-selects two-marks

Answer key

2.6

Graph Matching (1)

2.6.1 Graph Matching: GATE CSE 2003 | Question: 36



How many perfect matching are there in a complete graph of 6 vertices?

- A. 15
- B. 24
- C. 30
- D. 60

gatecse-2003 graph-theory graph-matching normal

Answer key

2.7.1 Graph Planarity: GATE CSE 1987 | Question: 2e



State whether the following statement is TRUE or FALSE:

There is a linear-time algorithm for testing the planarity of finite graphs.

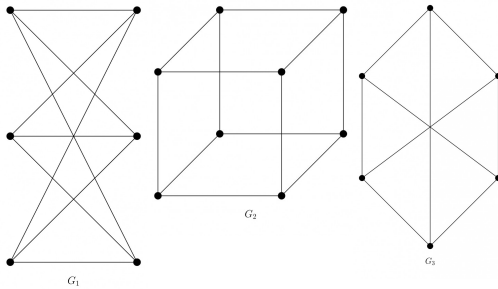
gate1987 graph-theory graph-planarity true-false

Answer key

2.7.2 Graph Planarity: GATE CSE 1989 | Question: 3-vi



Which of the following graphs is/are planar?



gate1989 normal graph-theory graph-planarity descriptive

Answer key

2.7.3 Graph Planarity: GATE CSE 1990 | Question: 3-xi



A graph is planar if and only if,

- A. It does not contain a subgraph homeomorphic to K_5 and $K_{3,3}$.
- B. It does not contain a subgraph isomorphic to K_5 and $K_{3,3}$.
- C. It does not contain a subgraph isomorphic to K_5 or $K_{3,3}$.
- D. It does not contain a subgraph homeomorphic to K_5 or $K_{3,3}$.

gate1990 normal graph-theory graph-planarity multiple-selects

Answer key

2.7.4 Graph Planarity: GATE CSE 1992 | Question: 01,x



Maximum number of edges in a planar graph with n vertices is _____

gate1992 graph-theory graph-planarity easy fill-in-the-blanks

Answer key

2.7.5 Graph Planarity: GATE CSE 1992 | Question: 02,viii



A non-planar graph with minimum number of vertices has

- A. 9 edges, 6 vertices
- B. 6 edges, 4 vertices
- C. 10 edges, 5 vertices
- D. 9 edges, 5 vertices

gate1992 graph-theory normal graph-planarity

Answer key

2.7.6 Graph Planarity: GATE CSE 2005 | Question: 10



Let G be a simple connected planar graph with 13 vertices and 19 edges. Then, the number of faces in the planar embedding of the graph is:

- A. 6
- B. 8
- C. 9
- D. 13

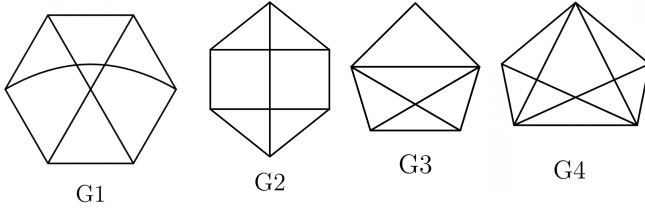
gatecse-2005 graph-theory graph-planarity

Answer key

2.7.7 Graph Planarity: GATE CSE 2005 | Question: 47



Which one of the following graphs is NOT planar?



- A. G1 B. G2 C. G3 D. G4

gatecse-2005 graph-theory graph-planarity normal

Answer key

2.7.8 Graph Planarity: GATE CSE 2008 | Question: 23



Which of the following statements is true for every planar graph on n vertices?

- A. The graph is connected B. The graph is Eulerian
C. The graph has a vertex-cover of size at most $\frac{3n}{4}$ D. The graph has an independent set of size at least $\frac{n}{3}$

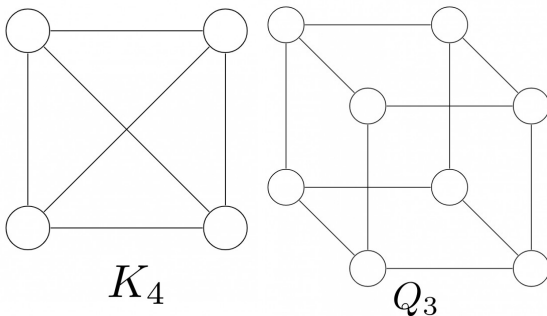
gatecse-2008 graph-theory normal graph-planarity

Answer key

2.7.9 Graph Planarity: GATE CSE 2011 | Question: 17



K_4 and Q_3 are graphs with the following structures.



Which one of the following statements is **TRUE** in relation to these graphs?

- A. K_4 is a planar while Q_3 is not B. Both K_4 and Q_3 are planar
C. Q_3 is planar while K_4 is not D. Neither K_4 nor Q_3 is planar

gatecse-2011 graph-theory graph-planarity normal

Answer key

2.7.10 Graph Planarity: GATE CSE 2012 | Question: 17



Let G be a simple undirected planar graph on 10 vertices with 15 edges. If G is a connected graph, then the number of **bounded** faces in any embedding of G on the plane is equal to

- A. 3 B. 4 C. 5 D. 6

gatecse-2012 graph-theory graph-planarity normal

Answer key

2.7.11 Graph Planarity: GATE CSE 2014 Set 3 | Question: 52



Let δ denote the minimum degree of a vertex in a graph. For all planar graphs on n vertices with $\delta \geq 3$, which one of the following is **TRUE**?

- A. In any planar embedding, the number of faces is at least $\frac{n}{2} + 2$
B. In any planar embedding, the number of faces is less than $\frac{n}{2} + 2$

- C. There is a planar embedding in which the number of faces is less than $\frac{n}{2} + 2$
D. There is a planar embedding in which the number of faces is at most $\frac{n}{\delta+1}$

gatecse-2014-set3 graph-theory graph-planarity normal

Answer key

2.7.12 Graph Planarity: GATE CSE 2015 Set 1 | Question: 54



Let G be a connected planar graph with 10 vertices. If the number of edges on each face is three, then the number of edges in G is _____.

gatecse-2015-set1 graph-theory graph-connectivity normal graph-planarity numerical-answers

Answer key

2.7.13 Graph Planarity: GATE CSE 2021 Set 1 | Question: 16



In an undirected connected planar graph G , there are eight vertices and five faces. The number of edges in G is _____.

gatecse-2021-set1 graph-theory graph-planarity numerical-answers easy one-mark

Answer key

Answer Keys

2.1.1	D	2.1.2	D	2.1.3	X	2.2.1	N/A	2.2.2	N/A
2.2.3	N/A	2.2.4	D	2.2.5	C	2.2.6	C	2.2.7	B
2.2.8	C	2.2.9	D	2.2.10	C	2.2.11	C	2.2.12	16
2.3.1	D	2.3.2	C	2.3.3	A	2.3.4	4	2.3.5	3
2.3.6	7	2.3.7	A;B	2.3.8	B;C	2.3.9	2	2.3.10	C
2.3.11	B	2.4.1	N/A	2.4.2	N/A	2.4.3	N/A	2.4.4	N/A
2.4.5	N/A	2.4.6	C;D	2.4.7	C	2.4.8	N/A	2.4.9	N/A
2.4.10	B	2.4.11	D	2.4.12	N/A	2.4.13	B	2.4.14	C
2.4.15	B	2.4.16	A	2.4.17	B	2.4.18	C	2.4.19	D
2.4.20	A	2.4.21	506	2.4.22	36	2.4.23	C	2.4.24	B
2.4.25	C;D	2.4.26	C	2.4.27	A	2.4.28	36	2.4.29	D
2.4.30	A	2.4.31	16	2.4.32	D	2.4.33	B	2.4.34	D
2.4.35	A	2.4.36	B	2.4.37	D	2.4.38	C	2.4.39	D
2.4.40	C	2.5.1	B	2.5.2	5	2.5.3	A;B;C	2.6.1	A
2.7.1	Q-Q	2.7.2	N/A	2.7.3	D	2.7.4	N/A	2.7.5	C
2.7.6	B	2.7.7	A	2.7.8	C	2.7.9	B	2.7.10	D
2.7.11	A	2.7.12	24	2.7.13	11 : 11				